Lab 8: Docker, containers, and images

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Objective

Containerize the Lab 5 Flask user API, build two distinct images, run them on ports 5000 and 3000, add a GET endpoint that returns all user names, and document Docker commands and results.

Background

A Dockerfile specifies how an image is built. An image is the packaged application and runtime; a container is a running or stopped instance of that image. Port publishing maps a host port to the port on which the application listens inside the container. Each container in this lab uses its own SQLite database.

Implementation

Version	Image tag	Behavior
1	js9872/lab8-joey_sade:0.0.1-RELEASE	Lab 5 user API; port 5000
2	js9872/lab8-joey_sade:0.0.2-RELEASE	Lab 5 API plus GET /api/get_users; port 3000

Docker permits only one colon in an image reference. The handout?s ?0.0.1:RELEASE? notation was therefore encoded as the valid tag 0.0.1-RELEASE, and likewise for version 2. The namespace js9872 follows the repository owner.

Build and run procedure

```
docker build -f Dockerfile.v1 -t js9872/lab8-joey_sade:0.0.1-RELEASE .
docker build -f Dockerfile.v2 -t js9872/lab8-joey_sade:0.0.2-RELEASE .
docker run --name lab8-v1 -p 5000:5000 -d js9872/lab8-joey_sade:0.0.1-RELEASE
docker run --name lab8-v2 -p 3000:3000 -d js9872/lab8-joey_sade:0.0.2-RELEASE
```

Both builds completed successfully. Both containers ran on the specified ports. Version 1 returned HTTP 200 from GET /api/users. For version 2, POST /api/users/add created a sample user, and GET /api/get_users returned ["Joey Sade"]. The API selects the name column from SQLite in user ID order.

Activity 1: Docker commands

docker ps -a --filter name=lab8

The -a flag includes running and stopped containers. The filter limits this snapshot to the two lab containers. The output shows both lab8-v1 and lab8-v2.

```
PS> docker ps -a --filter name=lab8
CONTAINER ID   IMAGE                                  COMMAND           CREATED          STATUS                       PORTS                                       NAMES
5ecf5e534cad   js9872/lab8-joey_sade:0.0.2-RELEASE   "python app.py"   46 seconds ago   Up 44 seconds                0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp   lab8-v2
bc52695a04ee   js9872/lab8-joey_sade:0.0.1-RELEASE   "python app.py"   47 seconds ago   Exited (137) 19 seconds ago                                              lab8-v1
```

Captured command output: docker ps -a --filter name=lab8

docker stop lab8-v2

Stops the named container and prints its name. Docker stop takes a container name or ID, not an image name.

```
PS> docker stop lab8-v2
lab8-v2
```

Captured command output: docker stop lab8-v2

docker logs lab8-v2

Prints standard output and error from the container. The captured Flask startup and HTTP request entries confirm the server ran and received requests.

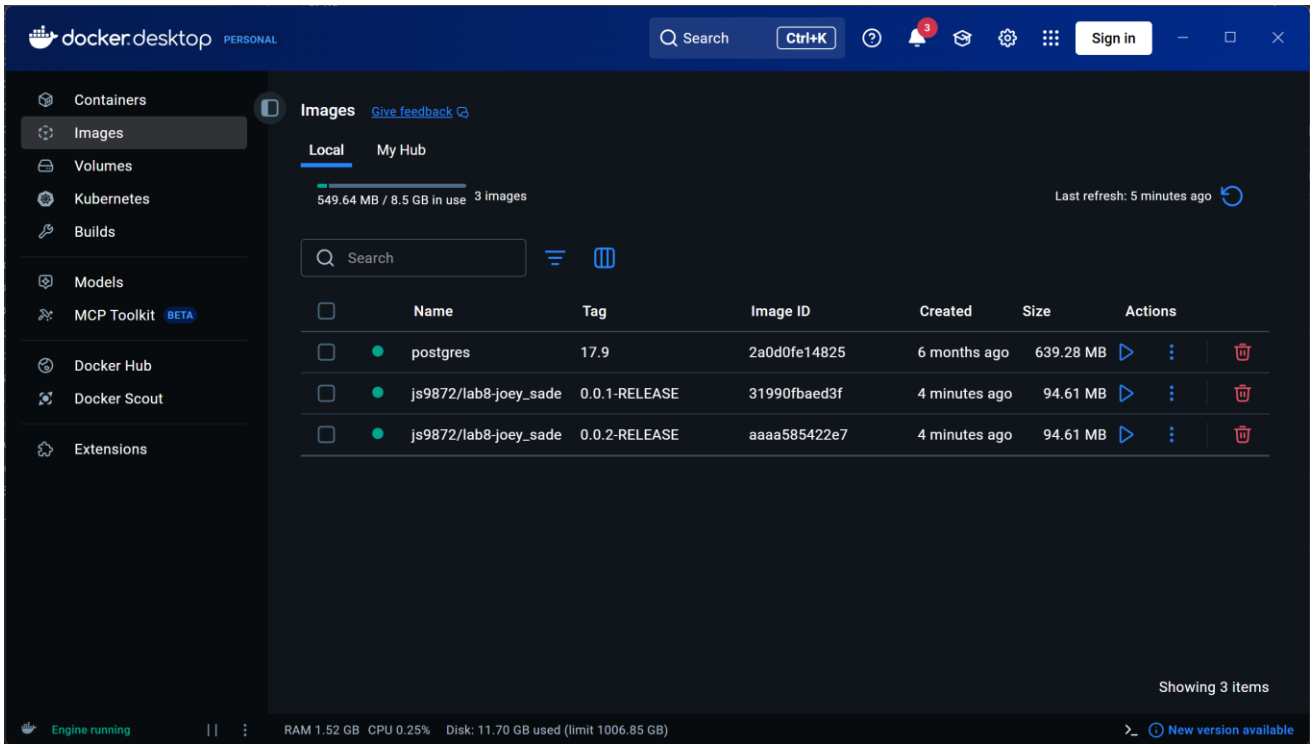
```
PS> docker logs lab8-v2
 * Serving Flask app 'app'
 * Debug mode: off
docker : WARNING: This is a development server. Do not use it in a production deployment. Use a production
WSGI server instead.
At line:2 char:214
+ ... oding utf8 evidence\docker-stop.txt; docker logs lab8-v2 2>&1 | Out-F ...
+                                           ~~~~~~~~~~~~~~~~~~~~~~~~~~~
    + CategoryInfo          : NotSpecified: (WARNIN...er instead.:String) [], RemoteException
    + FullyQualifiedErrorId : NativeCommandError

 * Running on all addresses (0.0.0.0)
 * Running on http://127.0.0.1:3000
 * Running on http://172.17.0.3:3000
Press CTRL+C to quit
172.17.0.1 - - [09/Nov/2026 12:30:09] "␛[35mPOST /api/users/add HTTP/1.1" 201 -
172.17.0.1 - - [09/Nov/2026 12:30:09] "GET /api/get_users HTTP/1.1" 200 -
```

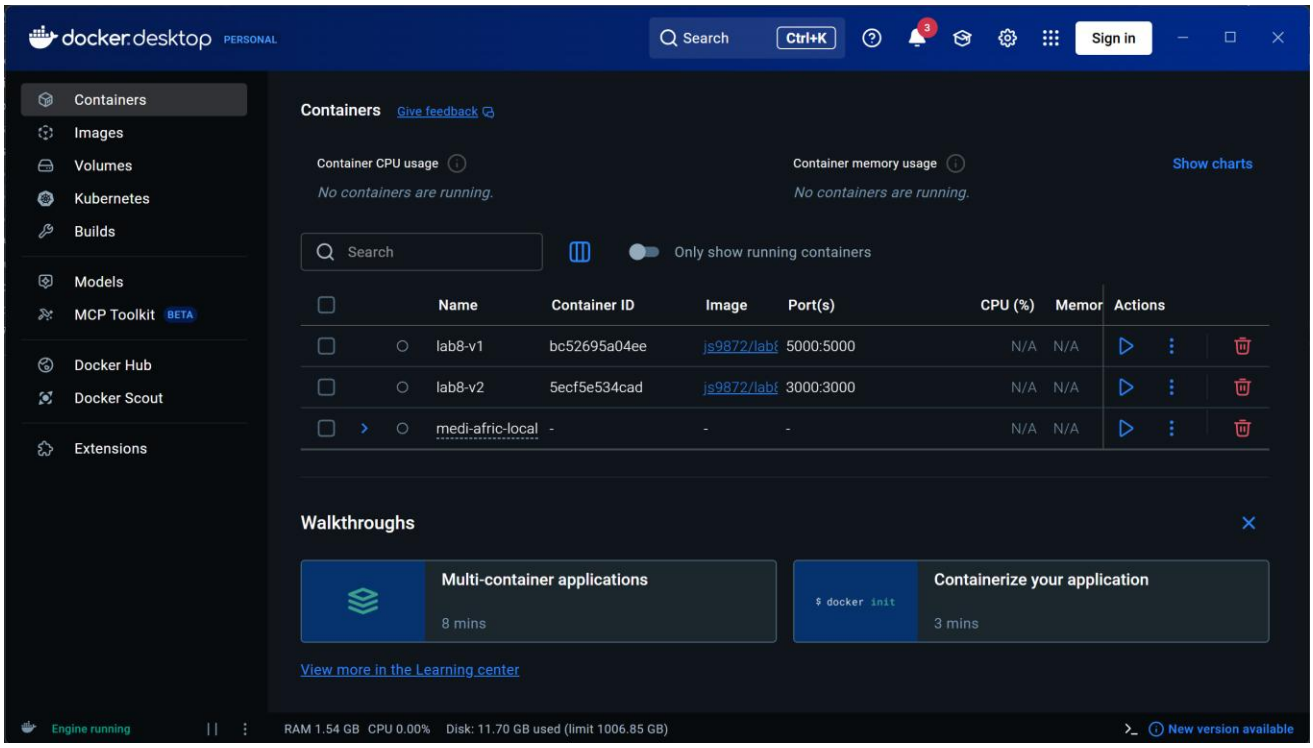
Captured command output: docker logs lab8-v2

Activity 2: two Docker images

Docker Desktop?s Images view shows both locally built image tags. Its Containers view shows the two named containers and their host-to-container port mappings after the stop commands. The stopped state is expected for the Activity 1 evidence.



Docker Desktop Images view: version 1 and version 2 tags.



Docker Desktop Containers view: lab8-v1 on 5000:5000 and lab8-v2 on 3000:3000.

```
PS> docker images js9872/lab8-joey_sade
REPOSITORY              TAG             IMAGE ID       CREATED              SIZE
js9872/lab8-joey_sade   0.0.2-RELEASE   aaaa585422e7   58 seconds ago       94.6MB
js9872/lab8-joey_sade   0.0.1-RELEASE   31990fbaed3f   About a minute ago   94.6MB
```

CLI image listing confirms both tags.

Verification and conclusion

The two image versions build and run locally. The first preserves the Lab 5 REST API on port 5000. The second adds the required name-list GET endpoint on port 3000; a sample insertion and retrieval confirmed the SQLite query. The command outputs and Docker Desktop captures document both graded activities. Source code, Dockerfiles, this report, and the evidence images are included in the submission ZIP.

Note: The AUB username was not present in the available project files, so the GitHub repository owner, js9872, was used as the image namespace.